All bivariate quartic tracial moment sequences admit size-two representations

Run `fa_banach_001`, lane 1 (GPT5.6)

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Status: candidate full result, likely valid. Every representable bivariate quartic real tracial moment sequence has a representing measure whose atoms have size at most two.

1 Source and target

For a normalized real tracial sequence $\beta = (\beta_w)_{|w|\leq 4}$ in two noncommuting self-adjoint variables, a representing measure is a finite convex combination

$$\beta_w = \sum_i \lambda_i \operatorname{Tr}\bigl(w(A_i, B_i)\bigr), \qquad \lambda_i > 0, \quad \sum_i \lambda_i = 1,$$

where A_i, B_i are real symmetric matrices of a common size within each atom and Tr is normalized trace. Its order-two moment matrix $\mathcal{M}_2$ is the 7×7 matrix indexed by $1, X, Y, X^2, XY, YX, Y^2$ [1].

Bhardwaj–Zalar prove that atoms of size at most two suffice in every case except a rank-six normal form satisfying $Y^2 = 1$ [1, pp. 3–4]. Their page-spanning statement is reproduced below.

The later paper [2] formulates the missing $Y^2 = 1$ assertion as its Conjecture 1. The previous full packet [3] proves that conjecture for arbitrary finite atom sizes, not merely for size-three atoms. The present note records the resulting all-rank conclusion.

then β admits a measure since all trace-positive polynomials of degree 4 are cyclically equivalent to sums of hermitian squares [BK10]. This is the duality established by [BK12, Theorem 4.4]. Moreover, the measure consists of at most 15 atoms of size 2 [BCKP13, Remark 3.9]. By dualizing a certain cone of trace-positive polynomials of degree 4 one obtains the so called completely positive semidefinite cone introduced by Laurent and Piovesan [LP15] which was used to study certain quantum graph parameters.

1.3. **Results.** In this paper we study the BQTMP for sequences with *singular* moment matrices. Initially, we approached this problem using a nc analog of the main tool for studying commutative sequences, i.e., finding rank preserving extensions of the moment matrices involved. As is already well established by Curto and Fialkow, the existence of a measure usually implies the existense of a rank preserving extension of a moment matrix, and hence a minimal measure with rank$(\mathcal{M}_2)$ atoms; see Theorem 2.7 below. However, in stark contrast to the commutative case, our research soon revealed that this does not apply for noncommutative sequences β. Characterizing moment matrices $\mathcal{M}_2$ which admit a flat extension is insufficient for solving the BQTMP. The most versatile tool in tackling the BQTMP is the application of appropriate affine linear transformations on the sequences β, see Subsection 2.5 below. This splits the BQTMP into finitely many cases according to the column relations that exist in $\mathcal{M}_2$. For $\mathcal{M}_2$ of rank at most 5, we completely characterize the existence, minimality and uniqueness of a measure in terms of the parameters β_w. In two out the four cases of $\mathcal{M}_2$ being of rank at most 6, we prove that the BQTMP is equivalent to the feasibility problem of certain linear matrix inequalities with an additional rank-to-variety condition for one of them. In all but a single case -

that of $\mathcal{M}_2$ being rank 6 and satisfying $\mathbb{Y}^2 = \mathbb{1}$ - we show that atoms of size at most 2 suffice in the minimal measure of β. We now give a brief outline of the results we prove in this paper.

Outline of the results on BQTMP: (We assume that β is a nc sequence.)

(1) If $\mathrm{rank}(\mathcal{M}_2) \leq 3$, then β does not admit a nc measure. Namely, if β admits a nc measure, then the columns $\mathbb{1}, \mathbb{X}, \mathbb{Y}, \mathbb{X}\mathbb{Y}$ in $\mathcal{M}_2$ must be linearly independent (see Corollary 2.3 below).
(2) If $\mathrm{rank}(\mathcal{M}_2) = 4$, then it suffices, after applying an appropriate affine linear transformation, to study the case when $\mathcal{M}_2$ satisfies the column relations

$$\mathbb{X}^2 = \mathbb{1}, \quad \mathbb{X}\mathbb{Y} + \mathbb{Y}\mathbb{X} = a\mathbb{1}, \quad \mathbb{Y}^2 = \mathbb{1},$$

where $a \in (-2, 2)$. By finding the representing atom of size 2 it turns out that such β always admits a nc measure. Moreover, the representing atom is unique (up to orthogonal equivalence); see Theorem 3.1.
(3) If $\mathrm{rank}(\mathcal{M}_2) = 5$, then it suffices, after applying an appropriate affine linear transformation, to study four special cases when $\mathcal{M}_2$ satisfies the column relation

Figure 1: The source identifies a single missing size-two case: rank six with $Y^2 = 1$ (arXiv:1611.00494, pp. 3–4).

2 Proof intuition

There are only seven rank possibilities for $\mathcal{M}_2$. The commutative case is already scalar: equality of the two cyclic fourth moments forces the commutator of every atom to vanish. For a noncommutative sequence the source forces rank at least four. Rank seven is the positive-definite theorem; ranks four and five are covered constructively in the source. At rank six, an invertible affine change of X, Y gives one of four quadratic normal forms. The source covers the first three, and the earlier GPT5.6 theorem covers the fourth, $Y^2 = 1$. Inverting the affine change does not alter matrix size.

3 Two elementary reductions

Lemma 1 (The commutative case is scalar)**.** *If a representable quartic tracial sequence satisfies* $\beta_{X^2Y^2} = \beta_{XYXY}$*, then it has a representing measure using only scalar atoms.*

Proof. For real symmetric matrices A, B, cyclicity of normalized trace gives

$$\operatorname{Tr}(A^2B^2) - \operatorname{Tr}(ABAB) = \frac{1}{2}\operatorname{Tr}\big((AB - BA)^T(AB - BA)\big) \geq 0.$$

Apply this identity to every atom of a representing measure and average with the positive densities. Equality of the two moments forces every summand on the right to vanish. Thus $A_iB_i = B_iA_i$ for every atom. Each real symmetric commuting pair is simultaneously orthogonally diagonalizable. Splitting its normalized trace over the joint eigenvectors replaces it by scalar atoms and preserves all moments (indeed, all word moments). ☐

Lemma 2 (Affine normalization preserves atom size)**.** *Let*

$$\widetilde{X} = a_0 + a_1X + a_2Y, \qquad \widetilde{Y} = b_0 + b_1X + b_2Y$$

with invertible linear part. A quartic sequence has a representation by atoms of size at most two if and only if its transformed sequence does.

Proof. Apply the displayed affine formulas to each pair (A_i, B_i), inserting the identity matrix for the constant terms. The matrix size is unchanged. The inverse affine formulas give the converse. ☐

4 Full theorem

Theorem 1. *Every normalized bivariate quartic real tracial moment sequence that admits a representing measure by finite real symmetric matrix pairs admits a representing measure whose atoms all have size at most two.*

Proof. Let $r = \operatorname{rank}\mathcal{M}_2 \leq 7$. Lemma 1 settles the commutative case, so suppose the sequence is noncommutative.

The source proves that $1, X, Y, XY$ are linearly independent in this case [1, Corollary 2.3]; hence $r \geq 4$.

If $r = 7$, then $\mathcal{M}_2$ is positive definite. The positive-definite quartic theorem gives a representing measure with at most fifteen atoms, all of size two [1, p. 3, citing Remark 3.9 of BCKP13].

If $r = 4$, Theorem 3.1 of [1] constructs a size-two atom after an invertible affine normalization. Lemma 2 transfers it back. If $r = 5$, the four normal forms are handled by Theorems 6.5, 6.8, 6.11 and 6.14 of [1]; their measures use scalar atoms and one size-two atom.

It remains that $r = 6$. Proposition 4.1(2) of [1], followed by an invertible affine change, reduces the sequence to one of

$$X^2 + Y^2 = 1, \qquad XY + YX = 0, \qquad Y^2 - X^2 = 1, \qquad Y^2 = 1.$$

The source proves that the first three normal forms have minimal measures with atoms of size at most two [1, p. 4]; see the evidence crop below.

$$\mathbb{X}\mathbb{Y} + \mathbb{Y}\mathbb{X} = \mathbf{0} \tag{1.6}$$

and one of

$$\mathbb{X}^2 + \mathbb{Y}^2 = \mathbb{1} \quad \text{or} \quad \mathbb{Y}^2 = \mathbb{1} \quad \text{or} \quad \mathbb{Y}^2 - \mathbb{X}^2 = \mathbb{1} \quad \text{or} \quad \mathbb{Y}^2 = \mathbb{X}^2 \tag{1.7}$$

(see Proposition 4.1 (1)). Due to the first relation the atoms in the minimal measure are of special form (see Proposition 5.1). In particular, all the atoms of size bigger than 1 have trace 0. Analyzing moment matrices $\mathcal{M}_2$ generated only by atoms of size bigger than 1 with trace 0, we show that only one such atom of size 2 is needed. Since every atom of size 2 generates a moment matrix of rank 4, β admits a nc measure if and only if there is a nc moment matrix $\widetilde{\mathcal{M}}_2$ of rank 4 representing a nc sequence $\widetilde{\beta}$ admitting a nc measure such that $\mathcal{M}_2 - \alpha\widetilde{\mathcal{M}}_2$, $\alpha > 0$, is a cm moment matrix admitting a measure. However, there are infinitely many possible atoms of size 2 satisfying relations (1.6) and (1.7). But there are at most 4 atoms of size 1 satisying (1.6) and (1.7). Therefore it is easier to subtract candidates for a cm moment matrices $\widehat{\mathcal{M}}_2$. Namely, we compute the smallest $\alpha > 0$ such that $\operatorname{rank}\left(\mathcal{M}_2 - \alpha\widehat{\mathcal{M}}_2\right) = 4$. By the solution of the rank 4 case we can characterize exactly, in terms of the parameters β_w, when a nc measure exists, type of a minimal measure and its uniqueness.

(4) If $\operatorname{rank}(\mathcal{M}_2) = 6$, then it suffices, after applying an appropriate affine linear transformation, to understand the four special cases when $\mathcal{M}_2$ satisfies one of the column relations

$$\mathbb{X}^2 + \mathbb{Y}^2 = \mathbb{1} \quad \text{or} \quad \mathbb{X}\mathbb{Y} + \mathbb{Y}\mathbb{X} = \mathbf{0} \quad \text{or} \quad \mathbb{Y}^2 - \mathbb{X}^2 = \mathbb{1} \quad \text{or} \quad \mathbb{Y}^2 = \mathbb{1}$$

(see Proposition 4.1 (2)). In the first three cases the atoms in the minimal measure for β are of size at most 2. In the first two cases the nc measure always exists if $\beta_X = \beta_Y = \beta_{X^3} = \beta_{X^2Y} = \beta_{Y^3} = 0$. This is proved by computing the smallest $\alpha > 0$ such that

$$\operatorname{rank}\left(\mathcal{M}_2 - \alpha\left(\mathcal{M}_2^{(1,0)} + \mathcal{M}_2^{(-1,0)}\right)\right) < 6$$

For the remaining form $Y^2 = 1$, Theorem 1 of the prior full packet [3] proves that every arbitrary representing measure can be replaced by one of type $(m, 1)$: one size-two atom and at most four scalar atoms. Finally use Lemma 2 to invert the normalization.

This exhausts $r = 4, 5, 6, 7$ and completes the proof. □

5 Verification and limitations

The reusable script `code/verify_rank_exhaustion.py` checks the commutator identity on 2,000 seeded random symmetric pairs of sizes one through nine and mechanically verifies that ranks one through seven and all four rank-six normal forms have an assigned branch. It is a regression test, not evidence for the cited source theorems.

This theorem is about degree-four moments in exactly two self-adjoint variables. It does not claim a size-two bound for higher truncation degree, more variables, complex non-self-adjoint variables, or prescribed supports.

The broad conjecture appears in commented source material of arXiv:1611.00494 rather than as a numbered statement in the final PDF. The published PDF nevertheless explicitly says that size two is proved "in all but a single case" and names that case as rank six with $Y^2 = 1$. The later paper [2] explicitly promotes that missing case to Conjecture 1.

6 Novelty check and review recommendation

On 11 August 2026 we searched the run registry and solution/attempt/proof-gap indexes for arXiv:1611.00494, the exact title, "bivariate quartic tracial moment", "atoms of size at most 2", and the $Y^2 = 1$ normal form. We also searched primary arXiv results for the exact conjecture and close atom-size phrases. We found [1], [2], the earlier run packet [3], and no later paper stating the all-rank synthesis. Novelty confidence is moderate because the conclusion is a short but apparently unrecorded corollary of those inputs.

Human review should focus on two interfaces: that Proposition 4.1 uses an invertible affine normalization, and that the source's "first three cases" are exactly the three displayed rank-six forms other than $Y^2 = 1$. The last form is the exact scope of Theorem 1 in [3].

References

[1] A. Bhardwaj and A. Zalar, *The singular bivariate quartic tracial moment problem*, arXiv:1611.00494, 2017.

[2] A. Bhardwaj and A. Zalar, *The tracial moment problem on quadratic varieties*, arXiv:2001.11614; J. Math. Anal. Appl. 498 (2021), 124936.

[3] Run `fa_banach_001`, *One size-two atom suffices for the quartic tracial moment problem on* $Y^2 = 1$, full solution packet for arXiv:2001.11614, GPT5.6, 2026.